

- ◆ The route being travelled by the vehicle and the person are pre-entered.
- ◆ Now, when the blind person approaches a crossing, the system will alert both, the car and the blind person. In this way, both will take extra precautions while the blind person crosses the road.
- ◆ This system could also help when small children with GPS devices attached cross the road by informing the drivers of kids in the vicinity.

From a technical point of view...

Currently, the challenge of detecting vehicles around remains. There are many 'Image Processing' algorithms that detect objects. All they need to be taught is how to detect a car.

But again, this approach has some disadvantages:

- ◆ Image processing algorithms require very high computational capability to carry out detections.
- ◆ When the blind need to navigate, this needs to happen very fast with images appearing at 25-30 frames per second (fps), something that's called 'real time' in the technology world.
- ◆ Existing Arduino™ boards don't have sufficient speed and capability to handle such demanding loads.

This is where the role of advanced devices with increased capabilities will come into play.



Detecting cars using an Image Processing algorithm

Another application area

Such devices can help in other application areas such as Robotics, where there's need for mobile robots to navigate a given space and perform some tasks. A camera mounted on top of the mobile robot can help detect objects and perform actions such as picking them up and placing them somewhere.

Maybe very soon this technology will be in our homes cleaning them.

The boards

Three such boards with higher capabilities are currently available – two are manufactured by Arduino™ along with partners and one is manufactured by Intel.